Towards Better Deepfake Detection

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***Abstract*—This document herein is a requirement towards fulfilling the criteria of participating in IEEE SPCUP 2025 held by ICASSP. Problem statement of the competition is deep fake detection of images. This document describes the approaches we evaluated and final solution from our team. We also highlight some interesting observations in this regard.**

***Index Terms*—Deepfake detection, transformers, binary classification**

I. Introduction

Though deepfakes have enabled creative applications such as in filmmaking, gaming, and personalized content creation, it has also opened the door to misuse in areas like misinformation campaigns, identity theft, and cyberbullying. Deepfakes have been a stagnant problem in the contemporary world. Deepfakes blend actual data with AI-generated content to deceive viewers into believing fake narratives. This evolving problem is not only a technical challenge but also a societal concern, with implications that extend into social, political, and ethical domains.

The given dataset consists of 219K fake images and 42K real images for training and 1.5K images in each category for validation. The dataset is generated through random selection from standard deepfake datasets like Celeb-DF-v1 [1], Celeb-DF-v2 [1], FaceForensics++ [2], DeepfakeDetection, FaceShifter [4], UADFV [6], Deepfake Detection Challenge Preview [5], and Deepfake Detection Challenge [3].

At the heart of deepfake detection lies the challenge of identifying subtle patterns or artifacts that distinguish synthetic media from authentic ones, regardless of the technique used to generate them. While early methods often struggled with generalization across datasets, recent advancements in universal detectors offer promise by leveraging imperceptible artifacts inherent to AI-generated content. This ongoing evolution highlights the need for interdisciplinary efforts, combining technical innovation with a broader understanding of societal impacts, to effectively tackle the growing threats posed by deepfakes. This paper focuses on developing a highly accurate and robust deepfake detection system, capable of tackling these challenges head-on.

Section II explores the related work in deepfake detection domain. Section III shows all the models and strategies we used to train our models. Section IV dives into finer details of our implementation. Section V shows the results with the suggested evaluation metrics.

II. Related Work

The ResNet-based approach in [29] is a pioneering effort in deepfake detection. It introduces a universal detector for distinguishing real and CNN-generated images, trained on synthetic data from 11 models like ProGAN [30], StyleGAN [31], and DeepFake. The study shows that classifiers trained on a single generator (e.g., ProGAN [30]) generalize well to unseen architectures, aided by detectable artifacts common to CNN-generated images. Data augmentation techniques, such as JPEG compression and Gaussian blur, further enhance robustness. This work sets a foundational benchmark in deepfake forensics, highlighting the current limitations of image synthesis.

The Xception [20] backbone leverages depthwise separable convolutions for efficient feature extraction. With 36 convolutional layers, it balances computational efficiency and performance. Fine-tuning on datasets like FaceForensics++ [2] and Celeb-DF [1] enhances its accuracy in detecting manipulated media. DSP-FWA [21] identifies face-warping artifacts using a ResNet-50 [22] model, while FFD [23] localizes manipulations

through attention mechanisms. CORE [24] enforces representation consistency via cosine distance regularization, and UCF [25] improves generalization by disentangling essential features. F3Net [26] extracts frequency-aware clues using cross-attention networks, and SPSL [27] detects up-sampling artifacts through spatial images and phase spectra. SRM [28] enhances generalization by combining high-frequency noise features from RGB and frequency domains, showcasing the framework's robustness in combating deepfakes.

III. Our Approach

We train our models in a supervised fashion using the given data. We trained individual models ranging from CNNs to model soups of transformers. Among vision transformers, we trained ViT [10], Swin transformer [15], ViT trained with initial weights of DINO [16] & MAE [17]. Respective ViTs reached 85.35% & 86.9% validation accuracy whereas the other models reached 90% with minimal tuning of hyperparameters. This motivated us to focus more on supervised checkpoints for initialization rather than unsupervised pretrained checkpoints. We realized we need to look for global attention based models. Along this line of thought, we found Swin Transformer [15] to be a good alternative, as it has shifted window attention which is similar to global attention. MaxViT [7] is another such model, which also has global attention-like modules. We also conducted experiments on CNN based models like ResNet [12], InceptionNet [11], [32], [33], EfficientNet [13]. However, these models' performance plateaued at about 94%. This motivated us to look at ConvNeXT [14] which is reminiscent of both transformers and CNNs. ConvNeXT tiny gave us the best performance of all models

On usage of foundational models: We used some VLMs like gpt-4o [19] and gpt-4o-mini [19]. GPT-4o [19] rejected most of the image prompts. However, gpt-4o-mini [19] rejected only 5 images. But we modified the prompt and it subsequently gave a clear response even for those images. However, Table I shows that gpt-4o-mini is nowhere near the task-specific trained models. A clear visualization of our pipeline can be seen in Figure 1.

IV. Implementation Details

Every model we tried is initialized with its ImageNet [18] trained weights. MaxViT [7] is trained with AdamW [9] optimizer with a cosine decay schedule where the learning rate is decreased from 10^{-4} to 10^{-5} over 20 epochs with a batch size of 64. RandAugment [8] with a magnitude of 15 and random horizontal flipping with a probability of 0.25 are used as data augmentation techniques. Swin transformer is trained with AdamW [9] optimizer with learning rate $5 \cdot 10^{-5}$ and a weight decay of 10^{-4}. ConvNext is trained with AdamW [9] optimizer with a learning rate of $5 \cdot 10^{-5}$ and a weight decay of 10^{-4}. Swin Transformer [15] is also trained with AdamW [9] optimizer for 10 epochs with a learning rate of $5 \cdot 10^{-5}$ and a weight decay of 10^{-4} with a batch size of 32. We used the cross entropy loss for training every model, apart from the Swin Transformer, where we used a balanced (weighted) cross entropy loss. The loss is scaled by $\frac{1}{219}$ for fake images and by $\frac{1}{42}$ for real images. These weights correspond to the inverse ratio of the number of fake and real images in the training dataset. This is being done to correct the bias due to a larger number of fake images present in the dataset.

A. Data Augmentation

During training, we apply various augmentations to introduce diverse variations into the data, enhancing the model's robustness and generalization. We use horizontal flipping with a probability of 0.5, random rotations within -10 to 10 degrees, Gaussian blur with a probability of 0.3 and a kernel size of 3 to 5, brightness and contrast adjustments applied with a probability of 0.5. These augmentations are applied based on an **augment probability**, determining the likelihood of applying them to a sample.

We initially set the augmentation probability to 0.3 and observed that the highest accuracy achieved by an individual model was 96.28%, specifically with ConvNeXT small. Increasing the augmentation probability to 0.5 did not result in significant changes. To address the imbalance in the training dataset, where real images were considerably fewer than fake images, we increased the augmentation probability for real images to 0.4 or 0.5 while keeping the probability for fake images unchanged at 0.3. This adjustment led to noticeable improvements, with the accuracy of ConvNeXT small rising to **96.97%**. Overall, incorporating these augmentations resulted in an accuracy improvement of 0.5-1%pt. These data augmentation techniques are applied to ConvNeXT [14] and Swin Transformer [15] .

V. Evaluation and Results

A. Metrics

We evaluate our models using standard binary classification metrics like accuracy, precision, recall, F1 score, AUC and equal error rate. Here, the negative class is for fake images and the positive class is for real images. If we denote true positives, true negatives, false positives, false negatives in the predictions by TP, TN, FP, FN respectively

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1 score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

$$= \frac{2TP}{2TP + FP + FN} \tag{5}$$

AUC is the area under the ROC curve, which is a plot of the recall, also known as the true positive rate (TPR),

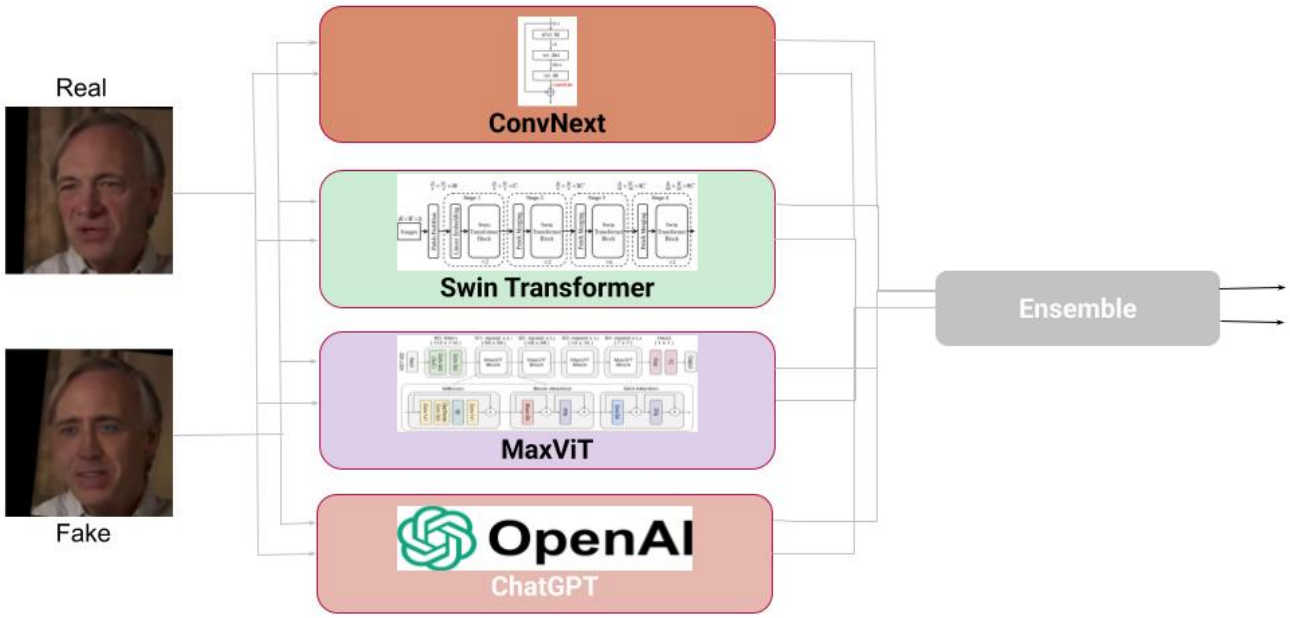


Fig. 1: A visualization of our prediction pipeline. This figure shows Ensemble 2.

Model	#Parameters	Time (ms)	Accuracy	AUC	F1-Score	Precision	Recall	EER
GPT-4o-mini	-	-	79.94	79.82	82.86	72.77	96.18	0.365
Swin Transformer	87.7M	28.8	96.02	96.02	96.08	95.58	96.57	0.045
MaxViT tiny	30.9M	32.9	96.09	96.09	96.12	96.06	96.18	0.040
ConvNeXT small	28.5M	7.22	96.97	96.95	97.05	95.32	98.83	0.049
Ensemble 1	**147.1M**	**71.2**	**97.40**	**98.80**	**97.45**	**96.05**	**98.90**	**0.039**
Ensemble 2	147M+	71.2+	97.49	98.78	96.12	96.06	96.18	0.036

TABLE I: Validation accuracy of various deepfake detection models. None of our trained models have non-trainable parameters. Time (ms) is benchmarked on NVIDIA RTX A5000 with a single image over multiple runs. #Parameters and time (ms) haven't been mentioned for GPT-4o-mini and Ensemble 2 owing to stochastic and closed source nature in OpenAI's services.

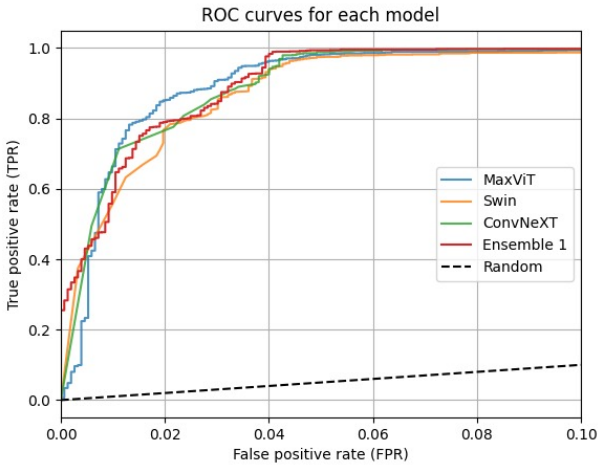


Fig. 2: Comparison of the ROC curves of different models along with our chosen ensemble 1. This is a **zoomed** version of the plot with the FPR varying in $[0, 0.1]$. Beyond 0.1 FPR, there is not much difference between the curves visually.

against the false positive rate (FPR) for different classification thresholds.

$$TPR = \frac{TP}{TP + FN} \tag{6}$$

$$FPR = \frac{FP}{TN + FP} \tag{7}$$

The equal error rate (EER) is the FPR corresponding to the point on the ROC curve where the false positive rate (FPR) and false negative rate (FNR) are equal.

$$FPR = FNR \tag{8}$$

$$= 1 - TPR \tag{9}$$

$$\text{i.e., } TPR + FPR = 1 \tag{10}$$

B. Ensemble Inference

We built ensembles using our best models with a soft voting strategy, i.e., the final ensemble prediction is a weighted sum of the confidence scores output by each individual model in the ensemble. These weights were optimized by performing a grid search on the validation dataset. Table I shows the scores achieved by each model and the built ensemble.

1) **Ensemble 1:** Swin Transformer base + MaxViT tiny + ConvNeXt small
2) **Ensemble 2:** Swin Transformer base + MaxViT tiny + ConvNeXt small + GPT-4o mini

Figure 1 shows the pipeline for Ensemble 2. Ensemble 1 is the same but without the GPT-4o predictions in the ensemble.

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